

HYUNG MU LEE

Hwaseong-si, Gyeonggi-do, South Korea | +82-10-7630-7085 | leehm1026@gmail.com | [LinkedIn](#)

ELECTRICAL ENGINEER

Highly dedicated and collaborative electrical engineer with a proven track record of expertise in high-speed interfaces, including PCIe, Ethernet, M-PHY, and more. My strong understanding of transceiver technology, coupled with expertise in signal and power integrity within state-of-the-art semiconductor processes, makes me a valuable asset in today's fast-paced technology landscape. Additionally, I have a comprehensive background in display systems, encompassing LCD, uOLED, and DDIC technologies, particularly in the context of VR systems for integration and validation. I excel at solving technical challenges, making cost-effective decisions, and offer strong communication and documentation skills. My collaborative nature enhances team performance.

Core Competencies

- Computer based Interface (APB, JTAG, SPI, I2C)
- SI / PI Equipment User
- PCB Module & Platform Designer
- MATLAB
- Python C Scripting
- Test Chip Application User

PROFESSIONAL EXPERIENCE

- Staff Electrical Engineer | Samsung, Hwasung-si, Gyeonggi-do** **04/2023 – Present**
- Verify and validate high-speed I/F chips implemented in advanced nano process technology.
 - Design FPGA-based system test PCBs and Develop test scripts utilizing Python and C.
 - Utilize various measuring and analyzing equipment such as UXR-series, JBERT, etc.
- Display Driver IC Engineer | Meta, Sunnyvale, CA** **05/2022 – 01/2023**
- Designed and integrated Display Driver ICs for LCD and uOLED systems.
 - Contributed to the development of high-speed interface (MIPI C/D-PHY) and power integrity modules.
 - Collaborated across departments, including OLED circuit, Optics, Process, and SoC teams.
- Display Electrical Engineer | Meta, Sunnyvale, CA** **02/2021 – 05/2022**
- Designed display driving platforms (PCBs) and scripts, successfully bringing up LCD panels.
 - Identified and resolved electrical issues, provided feature suggestions, and made informed decisions.
 - Managed multiple vendors for PCB design, panel design, and DDIC application.
- System Electrical Engineer | Facebook, Menlo Park, CA** **02/2019 - 01/2021**
- Designed MIPI high-speed display interface circuits.
 - Contributed to the mobile SoC power and thermal analysis efforts, including automation scripting.
 - Led the design and verification of new VR systems, leveraging the latest technology.
- Software Test Engineer | Verb Surgical, Mountain View, CA** **11/2018 – 02/2019**
- Worked on writing software verification protocols for firmware testing and executed the tests.
- Research Assistant | Johns Hopkins University, Baltimore, MD** **08/2016 – 08/2018**
- Conducted research on the Integrate and Fire Array Transceiver project, implemented in CMOS technology to build a biologically plausible model of visual information processing.
- Undergraduate Research Assistant | Konkuk University, Seoul, South Korea** **01/2014 – 11/2014**
- Designed an optical sensing circuit to obtain temperature and oxygen saturation signals from sensors.
- Software Engineering Intern | MyCityWay, New York, NY** **06/2013- 08/2013**
- IT Specialist | Republic of Korea Army, Kyunggi-do, South Korea** **03/2009 -01/2011**

FORMAL EDUCATION

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| Master of Science, Electrical and Computer Engineering | Johns Hopkins University, Baltimore, MD |
| Bachelor of Science, Electronics Engineering | Konkuk University, Seoul, South Korea |